Job Market Intelligence: Salary & Fake Post Prediction via Computational Methods

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Problem Statement

Rapid growth of online recruitment platforms has improved access to opportunities, but it has also increased salary inconsistency and exposure to fraudulent job listings. Candidates and recruiters need a reliable, data-driven intelligence system instead of manual verification and static rule-based checks.

- Online job portals face two major issues: **salary transparency gaps** and **fraudulent job advertisements**.
- Existing salary tools often miss key factors (skills, experience, role, geography), reducing reliability.
- Rule-based fake-post filters are less effective as scammers continuously change posting patterns.
- Users often rely on multiple disconnected tools, causing delays, confusion, and inconsistent outcomes.
- Need: one intelligent system that predicts fair salary and flags suspicious job postings in real time.

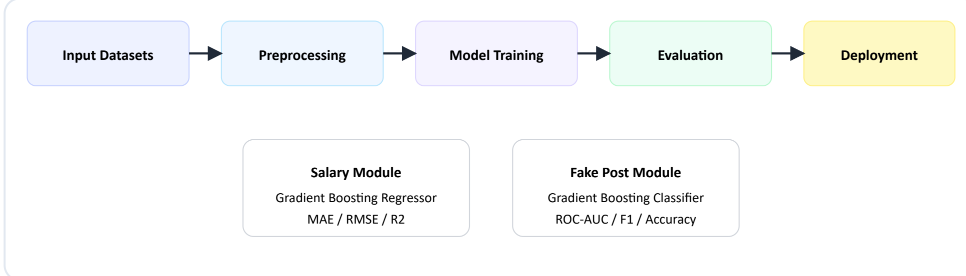
Objectives

- Develop a robust ML pipeline for salary prediction using role, skills, location, and experience features.
- Build an NLP/ML fake-job detection module to identify suspicious postings from text and URL inputs.
- Integrate both modules into a single interface to provide quick, actionable, and safe job-market decisions.
- Reduce uncertainty in salary negotiation through PPP-aware and data-driven compensation estimates.
- Improve user safety by maximizing fraud recall and minimizing missed scam listings.
- Enable trend-driven career planning through transparent outputs and interpretable model-driven insights.

Salary Accuracy Focus | Fraud Recall Focus | Unified Decision Support

Methodology / Approach

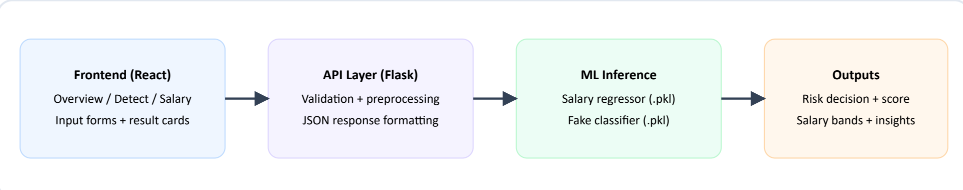
- **Workflow:** data collection → cleaning/encoding → model training → evaluation → deployment.
- **Algorithms:** Gradient Boosting Regressor for salary; Gradient Boosting Classifier for fake-post detection.
- **Validation:** train-test split with performance thresholds and repeatable benchmarking scripts.



Implementation Details

- **Tools/Tech:** Python, Flask API, React + Vite frontend, scikit-learn, pandas, numpy, joblib.
- **Data modules:** salary dataset processing, fake-post text/URL analysis, feature engineering.
- **Model modules:** regression training, classification training, comparative benchmarking scripts.
- **System modules:** backend APIs, dashboard UI, dataset stats explorer, prediction endpoints.

The backend exposes REST endpoints for salary prediction and fake-post classification. The frontend consumes these APIs and renders human-readable outputs (risk labels, confidence bars, salary ranges in INR/month-LPA/year) for practical decision support.



Frontend Interface Preview

Actual frontend screenshots from Major Project II (Overview, Fake Job detection, Salary prediction).

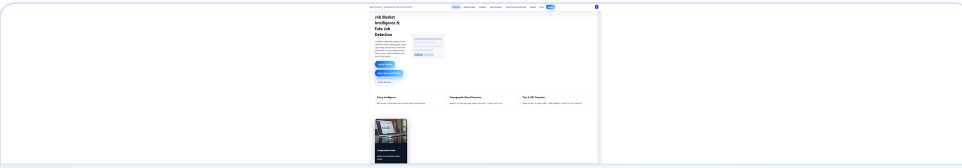


Fig A. Overview dashboard

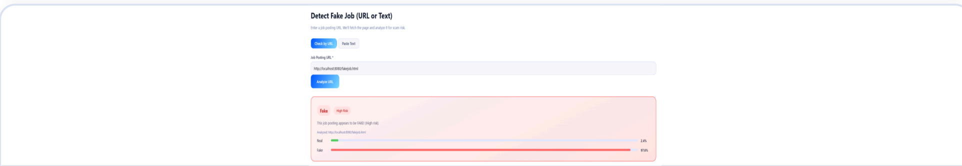


Fig B. Detect Fake Job (URL/Text)

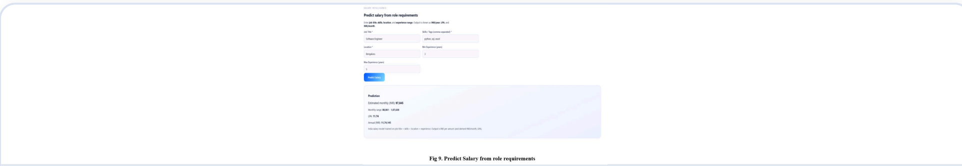
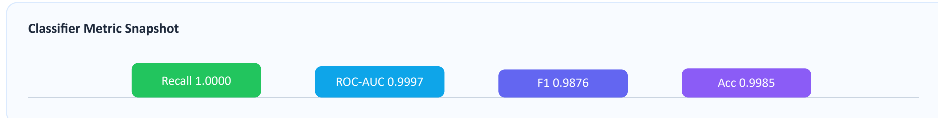


Fig C. Predict salary from role requirements

Results / Performance

- **Salary prediction:** MAE = 3,799; RMSE = 7,514; R2 = 0.977 (PPP-adjusted test set).
- **Fake-post detection:** ROC-AUC = 0.9997; Precision = 0.9754; Recall = 1.0000; F1 = 0.9876; Accuracy = 0.9985.
- Gradient Boosting consistently outperformed baseline alternatives in reliability and generalization.



Conclusion & Future Scope

Conclusion

- The project delivers one unified system for salary estimation and fake-post identification.
- Key objectives were achieved with strong quantitative performance and robust workflow design.
- Overall solution is effective for safer, data-driven job market decisions.

Future Scope

- Add explainable AI (SHAP/LIME), richer NLP embeddings, and advanced fraud signals.
- Scale as cloud-native microservices with real-time APIs and automated drift monitoring.
- Integrate with job portals/HR systems and support larger enterprise-level deployments.

Explainability | Cloud Scaling | Portal Integration

References (Top Selected)

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Publications (if applicable)

No publications associated with this work at present.